Ear — Hearin[illegible]fnes[illegible]

ENT | 3rd Year MBBS

Hi Sammy!
CSOM = most repeated
ENT topic in exams!
Let's nail it!

CSOM | Hearing Loss | ENT | 3rd Year

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Tuning fork tests are asked EVERY exam! Learn the table cold!

Parts of the Ear (Quick Recap)

Part	Structures	Function
External	Pinna, EAC, TM (lateral)	Collects & conducts sound
Middle	Ossicles, ET, mastoid	Impedance matching (x18)
Inner	Cochlea, vestibule, SCC	Transduction + balance

Types of Hearing Loss

Feature	Conductive (CHL)	Sensorineural (SNHL)
Site of lesion	External / Middle ear	Cochlea / CN VIII / central
Common causes	CSOM, otosclerosis, OM effusion	Presbycusis, ototoxicity, NIHL
Rinne test	NEGATIVE (BC > AC)	POSITIVE (AC > BC, both reduced)
Weber test	Lateralizes to DISEASED ear	Lateralizes to BETTER ear
ABC test	BC same both ears	BC reduced on affected side
Treatment	Often surgically correctable	Hearing aids, cochlear implant

Tuning Fork Tests (512 Hz)

Test	Method	Interpretation
Rinne	AC vs BC on same ear	+ve = AC>BC (normal/SNHL); -ve = BC>AC (CHL)
Weber	Fork on vertex/forehead	Lateralizes to diseased (CHL) / better (SNHL)
ABC	BC comparison both ears	Reduced on affected side = SNHL
Bing	Occlusion of EAC during BC	+ve = occlusion effect present (normal/SNHL)
Schwabach	Compare BC with examiner	Prolonged = CHL; Shortened = SNHL

★ **Beware: False Negative Rinne!**
In severe unilateral SNHL, BC from deaf ear is heard in normal ear (cross-hearing).
Use masking OR Stenger test to confirm malingering/unilateral severe SNHL.

Safe vs Unsafe CSOM = guaranteed LAQ!
This table is GOLD!

AOM vs CSOM

Feature	AOM (Acute)	CSOM (Chronic)
Duration	< 3 weeks	> 3 months (some say 6 wks)
TM	Congested / bulging / intact	Permanent perforation
Discharge	Purulent (if perforated)	Persistent / recurrent
Pain	Severe earache	Usually painless
Common in	Children (short, horizontal ET)	All ages
Organism	S. pneumoniae, H. influenzae	Pseudomonas, Proteus, S. aureus
Rx	Amoxicillin, analgesics	Aural toilet + drops +/- surgery

CSOM: Tubotympanic (Safe) vs Atticoantral (Unsafe)

Feature	Tubotympanic (Safe/Mucosal)	Atticoantral (Unsafe/Squamosal)
Site	Anteroinferior (pars tensa)	Posterosuperior / attic (pars flaccida)
Perforation	Central	Marginal / Attic
Discharge	Mucoid, non-foul	Foul-smelling, scanty, purulent
Cholesteatoma	Absent	PRESENT (hallmark!)
Granulations	Rare	Common (polyps)
Complications	Rare	Dangerous (intracranial!)
Treatment	Conservative first	ALWAYS surgical (mastoidectomy)
Hearing loss	Conductive (mild-mod)	Conductive / mixed

Cholesteatoma — Key Points

- Definition: Sac of keratinized squamous epithelium in middle ear / mastoid
- NOT a true tumor! It's a misnomer (no cholesterol, no tumor)
- Types: Congenital (behind intact TM) vs Acquired (primary/secondary)
- Acquired primary: Retraction pocket theory (most accepted) — Wittmaack
- Complications: Bone erosion ⊠ facial nerve palsy, labyrinthine fistula, brain abscess

Complications of CSOM (Atticoantral)

INTRACRANIAL

- Meningitis (most common IC)
- Brain abscess (temporal lobe / cerebellar)
- Lateral sinus thrombosis
- Otitic hydrocephalus
- Extradural abscess

EXTRACRANIAL

- Mastoid abscess (subperiosteal)
- Facial nerve palsy
- Labyrinthitis / labyrinthine fistula
- Petrositis (Gradenigo syndrome)
- Bezold abscess (SCM tip)

Investigations + Management

PTA + Tympanoplasty
types = guaranteed SAQ!
Know the table!

Investigations for Hearing Loss / CSOM

Investigation	What It Tells You
PTA (Pure Tone)	Type & degree of hearing loss (AC + BC thresholds)
Impedance / Tympanometry	Middle ear pressure, TM mobility, ET function
OAE	Outer hair cell function (neonatal screening)
BERA / ABR	Auditory pathway up to brainstem (objective test)
HRCT Temporal Bone	Bone erosion, cholesteatoma extent, complications
Pus C/S	Organism identification for directed antibiotic therapy

★ **PTA Quick Interpretation:**

AB gap present = Conductive HL | Both AC & BC reduced equally = SNHL
Degree: Mild (26-40 dB), Moderate (41-55), Mod-Severe (56-70), Severe (71-90), Profound (>90)
Normal hearing threshold = 0-25 dB | Socially adequate hearing = up to 40 dB

Medical Management of CSOM

Aural toilet:	Dry mopping / suction clearance under microscope
Topical drops:	Ciprofloxacin + Dexamethasone ear drops (1st line)
Systemic Abx:	Based on C/S; Amoxiclav / Cefixime for acute flare
Precautions:	Keep ear dry, avoid swimming, no ear buds

Surgical Management

Surgery	Indication & Details
Myringoplasty	Repair of TM perforation alone (dry ear, small perforation)
Tympanoplasty	TM repair + ossicular chain reconstruction (Wullstein types I-V)
Cortical Mastoidectomy	Safe CSOM with mastoiditis; preserve middle ear
Modified Radical	Unsafe CSOM; remove disease, preserve hearing mechanism
Radical Mastoidectomy	Extensive disease; make ear safe (hearing sacrificed)
Ossiculoplasty	Reconstruct eroded ossicles (PORP / TORP prosthesis)

Wullstein Classification of Tympanoplasty

Type	Graft Placed On	Clinical Use
Type I	TM only (= Myringoplasty)	Intact ossicular chain
Type II	Incus / remnant of incus	Eroded incus
Type III	Stapes head (columella)	Malleus + incus absent
Type IV	Stapes footplate	Only footplate remains

WHO classification of deafness is frequently asked in theory exams!

WHO Classification of Hearing Impairment

Grade	Hearing Level	Description
0 - Normal	< 25 dB	No or very slight hearing problems
1 - Slight	26-40 dB	Able to hear whispers; difficulty in soft speech
2 - Moderate	41-60 dB	Difficulty with normal speech at 1 meter
3 - Severe	61-80 dB	Can hear only shouted / amplified speech
4 - Profound	> 80 dB	Cannot hear even amplified speech; deaf

Causes of Deafness (Quick Classification)

Category	Conductive HL	Sensorineural HL
Congenital	Atresia of EAC, ossicular malform.	Rubella, CMV, genetic (Waardenburg)
Infective	CSOM, otitis externa, AOM	Measles, mumps, meningitis
Traumatic	TM perforation, ossicular disruption	Temporal bone fracture, noise-induced
Neoplastic	Glomus tumor, exostosis	Acoustic neuroma (CN VIII schwannoma)
Others	Otosclerosis, ET dysfunction	Presbycusis, ototoxicity, Meniere's

Ototoxic Drugs (High Yield!)

- Aminoglycosides (Streptomycin=vestibulotoxic, Neomycin=cochleotoxic) — MOST IMPORTANT
- Loop diuretics (Furosemide), Cisplatin, Quinine/Chloroquine, Aspirin (tinnitus)
- Mnemonic: 'PLAN C' = Platinum(cisplatin), Loop diuretics, Aminoglycosides, NSAIDs, Chloroquine

Rehabilitation of Deafness

Modality	Details
Hearing aids	BTE / ITE / CIC; for mild-severe HL (amplification)
Cochlear implant	For bilateral profound SNHL; bypasses cochlea to CN VIII
BAHA	Bone-anchored; for conductive / mixed HL, single-sided deafness
Auditory training	Speech therapy + lip reading + sign language
FM systems	Classroom amplification for children

★ NPPCD (National Programme for Prevention & Control of Deafness)

Launched: 2007 (pilot), scaled up 2012-13 | Target: Reduce deafness burden by 25%

Strategy: Early detection, treatment of ear diseases, rehab, awareness

India: ~6.3% population has disabling hearing loss (WHO estimates)

CSOM + Hearing Loss =
guaranteed 20+ marks!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Define CSOM. Classify. Describe features & management of safe vs unsafe types
- 2. Classify hearing loss. Describe tuning fork tests with interpretation
- 3. Describe cholesteatoma — types, pathogenesis, complications, surgical Rx
- 4. Enumerate complications of CSOM. Describe management of intracranial comps
- 5. Describe types of tympanoplasty (Wullstein). Indications for mastoidectomy

Most Repeated SAQs (5–6 marks)

- 1. Differentiate safe vs unsafe CSOM (table format)
- 2. Tuning fork tests — Rinne, Weber, ABC interpretation
- 3. Types of tympanoplasty (Wullstein classification)
- 4. Cholesteatoma — define, types, complications
- 5. Pure tone audiometry — principle and interpretation
- 6. Ototoxic drugs with examples

LAQ Answer Blueprint (CSOM)

1	Define	Chronic inflammation of middle ear cleft with TM perforation
2	Classify	Safe (tubotympanic) vs Unsafe (atticoantral) — draw table!
3	Features	Discharge, hearing loss, examination findings, TM findings
4	Investigations	PTA, impedance, HRCT temporal bone, pus C/S
5	Complications	Intracranial vs Extracranial (list both columns)
6	Management	Medical (aural toilet + drops) + Surgical (mastoidectomy types)

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Safe CSOM = which type?	Tubotympanic (mucosal)
Unsafe CSOM hallmark	Cholesteatoma
Most common complication of CSOM	Conductive hearing loss
Most common intracranial comp of CSOM	Meningitis
Foul-smelling ear discharge	Atticoantral CSOM (cholesteatoma)
Most common cause of CHL in children	Otitis media with effusion (glue ear)
Gradenigo syndrome triad	Otorrhea + retro-orbital pain + CN VI palsy
Brown sign (on otoscopy)	Glomus tumor (blanches on pneumatic otoscopy)
Otosclerosis: hearing loss type	Conductive (Carhart's notch at 2 kHz on PTA)
Schwartze sign	Otosclerosis (flamingo-pink TM)
Bilateral profound SNHL treatment	Cochlear implant
Most ototoxic aminoglycoside	Neomycin (cochlea); Streptomycin (vestibule)

♥ Every ear you treat gives someone the gift of hearing, Sammy!